

COVER SHEET, DRAWING INDEX & LEGEND

Electric batch oven - 18 kW heater bank, 2 x recirculation blowers, PLC / HMI control retrofit

1. DESIGN BASIS

Incoming supply	415 V AC, 3-phase + N + PE, 50 Hz, TN-S
Prospective Isc	10 kA rms sym. at panel incomer
Control supply	230 V AC ex 500 VA control transformer -T1
DC supply	24 V DC, 5 A regulated SMPS -G1
Heater load	18 kW total - 3 x 6 kW elements, star, 25 A/ph
Blower load	2 x 1.5 kW, 415 V, 3-ph, 3.5 A FLC, DOL start
Heater switching	Contactor -K3, driven from PLC output Y2
Temp. control	Panel PID controller -A2, relay output OUT1
	wired as a volt-free signal to PLC input X6
Over-temperature	Independent high-limit thermostat -B3 wired directly in the -K3 coil circuit (hard-wired trip)
Door interlock	Position switch -B1 to PLC input X5 only (software stop of blowers - see NOTE 4)
PLC	MELSEC iQ-F FX5U-32MT/ES (16 DI / 16 DO, sink)
HMI	GOT2000 GT2107-WTBD 7" TFT, 24 V DC, Ethernet
PLC <-> HMI link	Ethernet 100BASE-TX, MELSOFT connection
Door watchdog	Chatter or no 'door closed' signal for 10 min -> alarm + volt-free output Y12 / -KA4
Data logging	FX5U SD card, 7-day rolling history, CSV
Enclosure	Wall-mounted 800 x 600 x 250 mm, IP54, RAL 7035
Standards	IEC 60204-1, IEC 61439-1/-2, IS 8623

2. CONTROL FUNCTIONS IMPLEMENTED IN THE PLC

- a) Blower 1 / Blower 2 start-stop from panel push-buttons and from the HMI.
- b) Door interlock: blowers stop while the door is open, heater stays ON; blowers restart automatically when the door is closed again.
- c) Heater contactor -K3 follows the PID controller demand signal (X6), gated by PLC permissives (E-stop, overloads, high-limit, no fault).
- d) Heat-up watchdog alarm: the demand contact X6 stays closed continuously for longer than the preset time, i.e. the oven never reaches set point and the controller never drops out => 'FAILURE TO REACH SETPOINT'.
- e) Door-signal watchdog: a chattering door switch, or no 'door closed' signal for 10 minutes, raises an alarm and energises the volt-free remote alarm output Y12 / -KA4 on terminals -X1:50 / -X1:51.
- f) Six independent retentive 2-hour slot timers (slot 1...6). Once started a timer runs to 02:00:00 and can only then be reset. An earlier reset needs the maintenance password (HMI security level 2).
- g) Door open / door close event counters, separate and resettable.
- h) Lots loaded / lots unloaded counters derived from slot-timer start and reset events, with a mismatch indication.
- i) Password-protected manual test of the blowers and the heater (level 2).
- j) Alarm and event history logged to the FX5U SD card, 7 days rolling, exportable to a USB stick from the HMI.

6. REVISION HISTORY

REV	DATE	DESCRIPTION	BY
0	2026-09-06	First issue - schematic for review	CLAUDE
1	2026-09-06	Door-signal watchdog alarm output	CLAUDE
		Y12 / -KA4 added; single maintenance	
		password level; 7-day data logging to	
		the SD card in the FX5U	

3. DRAWING INDEX

SH.	DRAWING No.	TITLE
01	OVN-2026-01-01	Cover sheet, drawing index & legend
02	OVN-2026-01-02	Single line diagram - power distribution
03	OVN-2026-01-03	Power circuit - blower motors -M1 / -M2
04	OVN-2026-01-04	Power circuit - heater bank -E1/-E2/-E3
05	OVN-2026-01-05	Control supply, E-stop & 24 V DC distribution
06	OVN-2026-01-06	PLC -A1 : CPU supply & digital inputs X0-X17
07	OVN-2026-01-07	PLC -A1 : digital outputs Y0-Y17
08	OVN-2026-01-08	Contactor coil circuits & interlock provision
09	OVN-2026-01-09	Temperature controller & over-temp protection
10	OVN-2026-01-10	HMI -A3, Ethernet network & door layout
11	OVN-2026-01-11	Panel general arrangement & gland plate
12	OVN-2026-01-12	Terminal strip schedule -X1 / -X2
13	OVN-2026-01-13	I/O schedule & bill of materials

4. LEGEND (IEC 60617)

	Contact, normally open		Circuit breaker, thermal-magnetic
	Contact, normally closed		Switch-disconnector / isolator
	Push-button, NO, spring return		Contactor, main pole
	Push-button, NC, spring return		Thermal overload relay
	Emergency stop, latching, NC		Motor, 3-phase asynchronous
	Position (door) switch, roller		Resistive heating element
	Contactor / relay coil		Control transformer
	Indicator lamp, 24 V DC		AC/DC power supply unit
	Audible alarm (hooter)		Thermocouple, type K
	Fuse		Terminal / protective earth

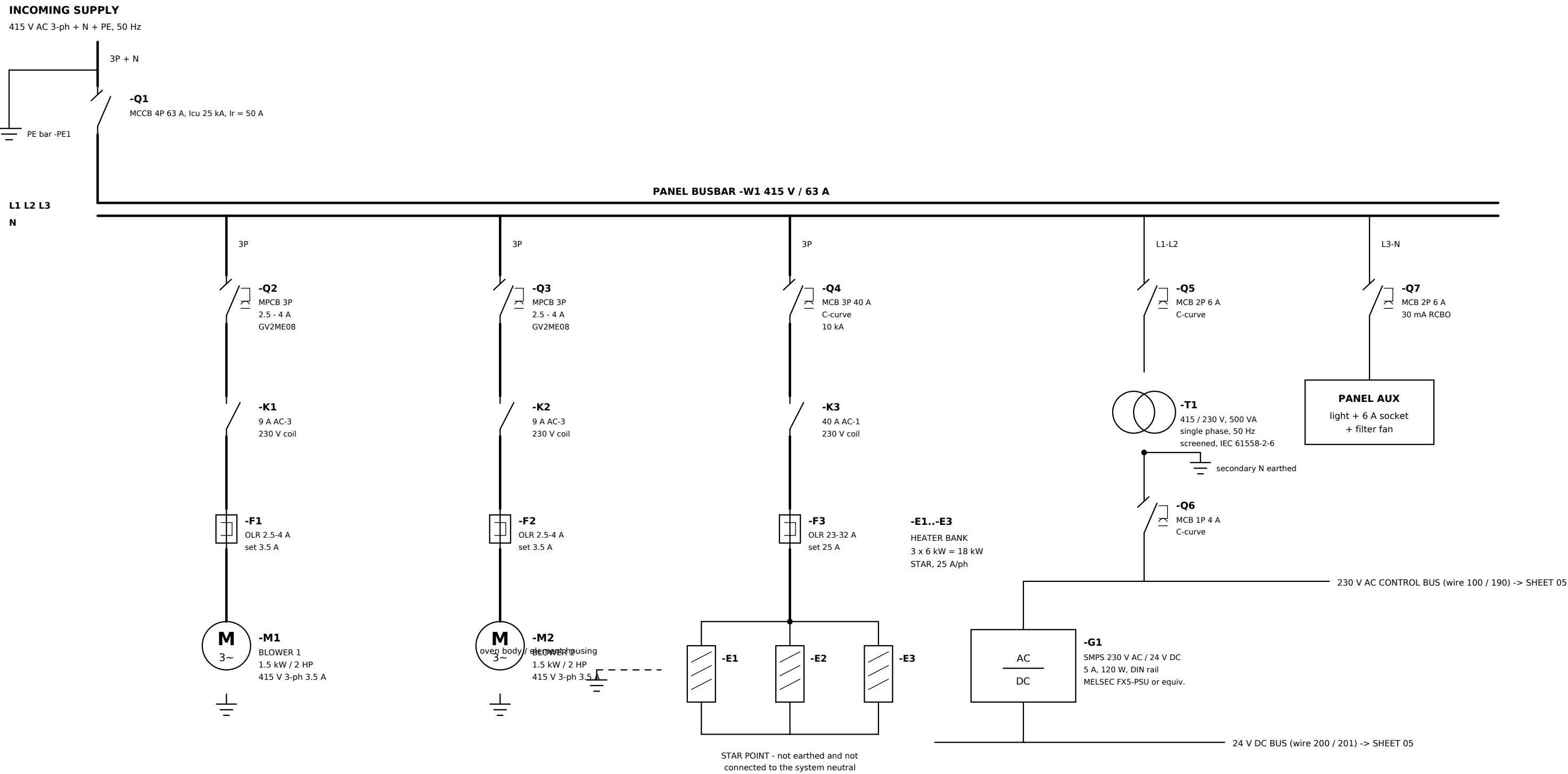
5. NOTES

1. All wiring to IEC 60204-1. Power conductors 1100 V grade PVC copper, control conductors 1.5 mm2 black, 24 V DC control conductors 1.0 mm2 dark blue, protective conductor green/yellow, neutral light blue.
2. All field wiring terminates on terminal strip -X1 / -X2. No field cable is landed directly on a device.
3. Heater over-temperature protection is HARD-WIRED: high-limit thermostat -B3 (manual reset) is in series with the -K3 coil, so the heater drops out on over-temperature regardless of the state of the PLC.
4. DOOR INTERLOCK - the door position switch -B1 is wired to PLC input X5 only, as instructed. The blowers are therefore stopped by the PLC program (software), not by hardware. A wire link -LK1 is fitted in the blower coil circuit on sheet 08; if a hard-wired interlock is required later, remove -LK1 and land the NC contact of a second (safety) door switch on terminals -X1:60 / -X1:61. No panel rework is needed.
5. The door interlock deliberately does NOT break the heater circuit: the heater stays energised while the door is open, per the process requirement. Warning label to be fitted at the door: 'HOT SURFACES - HEATER LIVE'.
6. PLC digital inputs are wired SINK type: terminal S/S is connected to +24 V and the field contact switches 0 V into the X terminal. PLC digital outputs are transistor SINK type: the load sits between +24 V and the Y terminal, COMO / COM1 are connected to 0 V. Contactor coils (230 V AC) are driven via interposing relays.
7. All fail-safe field contacts (E-stop, high-limit, stop push-buttons) are wired normally CLOSED so that a broken wire produces the safe state. Overload trip contacts are wired NO (input ON = tripped) for diagnostics.
8. 0 V of the 24 V DC system is bonded to PE at one point only, at terminal -X2:0V/PE in the panel.
9. Cable screens of the thermocouple extension lead are earthed at the panel end only.
10. Ratings shown are for the assumed loads listed in section 1. Confirm motor nameplate FLC and heater element rating on site and adjust -Q2/-Q3 settings, -F1/-F2/-F3 overload settings and cable sizes accordingly.

PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE COVER SHEET, DRAWING INDEX & LEGEND						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	01 / 13	1

SINGLE LINE DIAGRAM - POWER DISTRIBUTION

415 V AC, 3-ph + N + PE, 50 Hz, TN-S, 10 kA



NOTES:

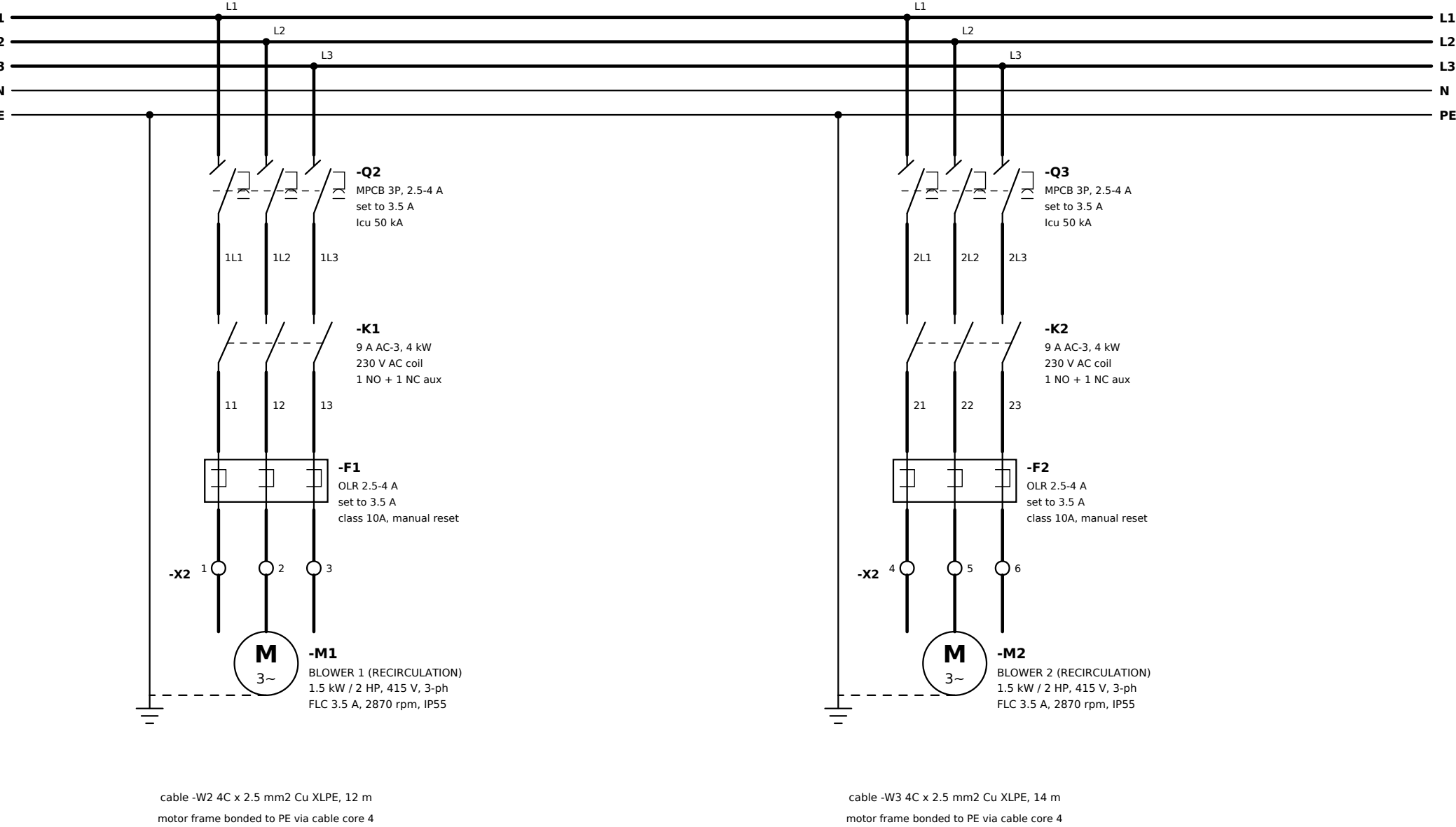
- 1. Diagram is single-line; all switching devices are 3-pole unless noted. See sheets 03 and 04 for the 3-line detail.
- 2. Heater elements are connected in STAR; the star point is not connected to the system neutral.
- 3. Cable schedule: -M1/-M2 4C x 2.5 mm2 Cu; heater bank 4C x 10 mm2 Cu; incomer 4C x 16 mm2 Cu.
- 4. -Q4 is a C-curve MCB sized for the resistive load; -F3 gives the overload/asymmetry protection for the elements.
- 5. -K3 is rated for AC-1 utilisation. Expect contact wear proportional to the PID cycle time - see NOTE 6 on sheet 04.

PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE SINGLE LINE DIAGRAM - POWER DISTRIBUTION						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	02 / 13	1

POWER CIRCUIT - BLOWER MOTORS -M1 / -M2

Direct-on-line starters, 415 V 3-ph, 1.5 kW each

FROM PANEL BUSBAR -W1 (SHEET 02) 415 V AC, 3-ph + N + PE, 50 Hz



AUXILIARY CONTACT ALLOCATION (for reference only)		
DEVICE	CONTACT	USED FOR
-K1	13-14 (NO)	PLC input X14 - run feedback sheet 06
-K1	21-22 (NC)	spare
-K2	13-14 (NO)	PLC input X15 - run feedback sheet 06
-K2	21-22 (NC)	spare
-F1	95-96 (NC)	in series with -K1 coil sheet 08
-F1	97-98 (NO)	PLC input X10 - OL tripped sheet 06
-F2	95-96 (NC)	in series with -K2 coil sheet 08
-F2	97-98 (NO)	PLC input X11 - OL tripped sheet 06
-Q2	aux 1 NO	spare (future MCB trip alarm)
-Q3	aux 1 NO	spare (future MCB trip alarm)

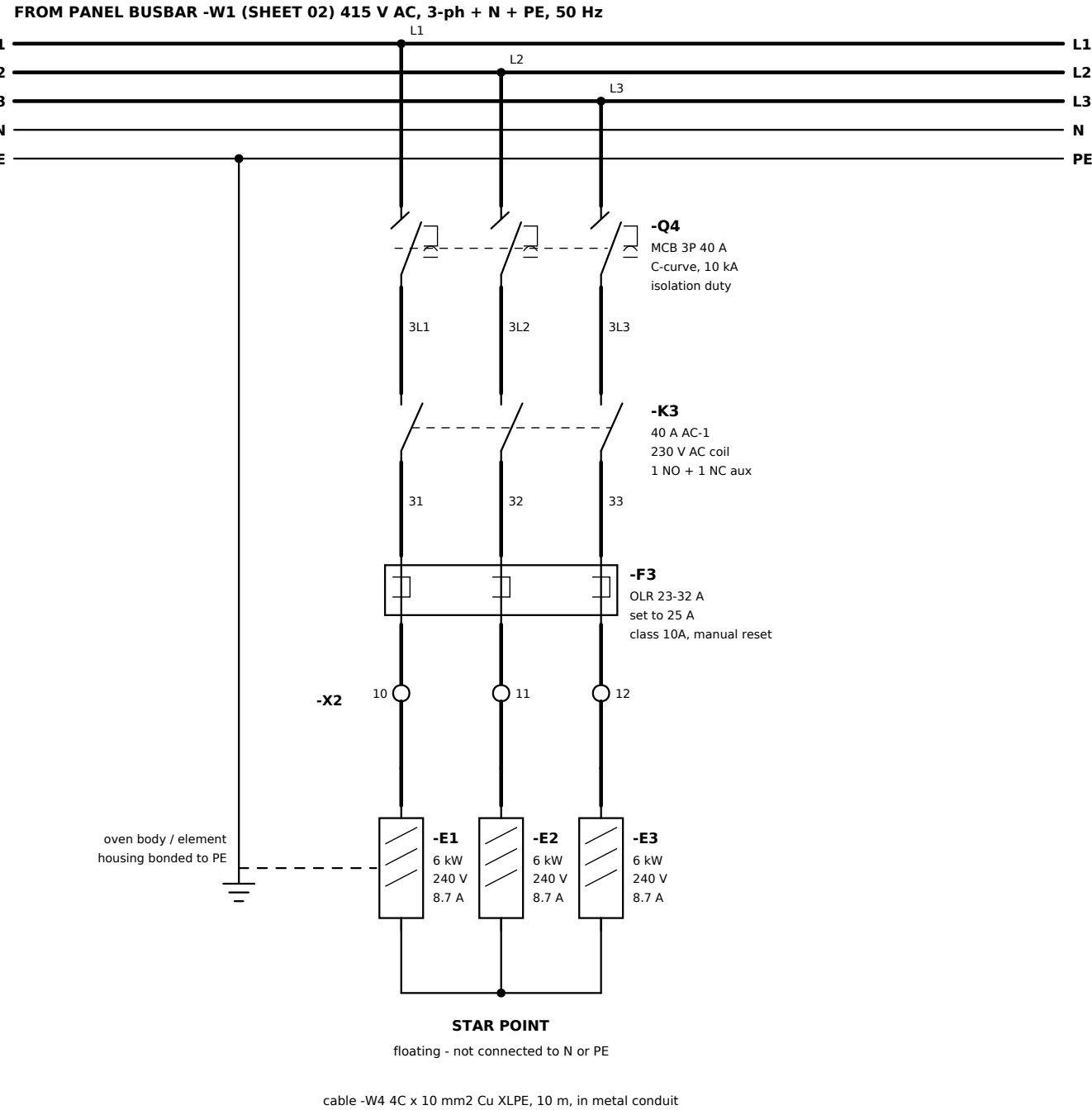
NOTES:

- Q2 / -Q3 are motor-protection circuit breakers providing short-circuit and isolation duty; the thermal overload relays -F1 / -F2 provide the running overload and phase-loss protection and give the trip signal to the PLC.
- Set both MPCB and OLR to the motor nameplate FLC after commissioning - the values shown assume 3.5 A.
- Overload relays are set to MANUAL RESET. A tripped overload must be reset at the panel; the PLC will not restart the blower until the trip input clears and the corresponding fault is acknowledged on the HMI.
- Both starters are direct-on-line. Motor direction to be checked at commissioning - airflow must be towards the oven plenum. Swap two phases at -X2 if rotation is incorrect.
- Blower contactors -K1 / -K2 are energised only when the PLC output Y0 / Y1 is ON - see sheets 07 and 08.

PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE POWER CIRCUIT - BLOWER MOTORS -M1 / -M2						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	03 / 13	1

POWER CIRCUIT - HEATER BANK -E1 / -E2 / -E3

3 x 6 kW resistive elements, star connected, 18 kW total, 25 A per phase



HEATER CIRCUIT CALCULATION

Total connected load	$P = 3 \times 6 \text{ kW} = 18 \text{ kW}$
Line current (star, 415 V)	$I = 18000 / (1.732 \times 415) = 25.0 \text{ A}$
Element voltage (star)	$U = 415 / 1.732 = 240 \text{ V}$
Element current	$I = 6000 / 240 = 25.0 \text{ A per element}$
Element resistance (cold)	$R = 240 / 25 = 9.6 \text{ ohm nominal}$
Cable sizing (25 A, 40 C)	10 mm2 Cu -> 57 A capacity, derated 0.87 $= 49 \text{ A} > 25 \text{ A OK}$
MCB -Q4	40 A C-curve $> 1.45 \times 25 \text{ A} = 36 \text{ A OK}$
Overload -F3	set 25 A, class 10A
Volt drop at 10 m	approx. 0.7 V (0.17 %) OK

CONTACTOR -K3 DUTY (see NOTE 6)

-K3 is switched by PLC output Y2, which follows the PID controller demand contact on input X6.

With a PID cycle time of 20 s the contactor would make and break up to 180 operations per hour, i.e. about 1.5 million operations per year on a 2-shift pattern.

The PLC program therefore imposes a minimum ON time and a minimum OFF time of 20 s each on Y2 (anti-chatter timers T_HTR_MIN_ON / T_HTR_MIN_OFF), which caps the switching rate at 90 operations per hour.

Set the controller cycle time to 20 s or longer to suit.

HEATER PROTECTION - LAYERS

- Q4 MCB 3P 40 A**
short-circuit protection and isolation of the heater feeder
- F3 thermal overload relay**
overload, element failure and phase asymmetry; manual reset
- B3 high-limit thermostat**
independent over-temperature trip, HARD-WIRED in the -K3 coil circuit (sheet 08) - it drops the heater even if the PLC or the PID controller fails. Manual reset at the panel.
- A2 PID controller**
closed-loop process temperature control; sensor-break alarm to X7
- PLC -A1**
start permissives, heat-up watchdog alarm and manual-test interlocks

IMPORTANT:

The door interlock does NOT break the heater circuit. The heater stays energised while the oven door is open, as required by the process. Layer 3 is then the only protection limiting the oven temperature, so -B3 must be function-tested at every planned maintenance.

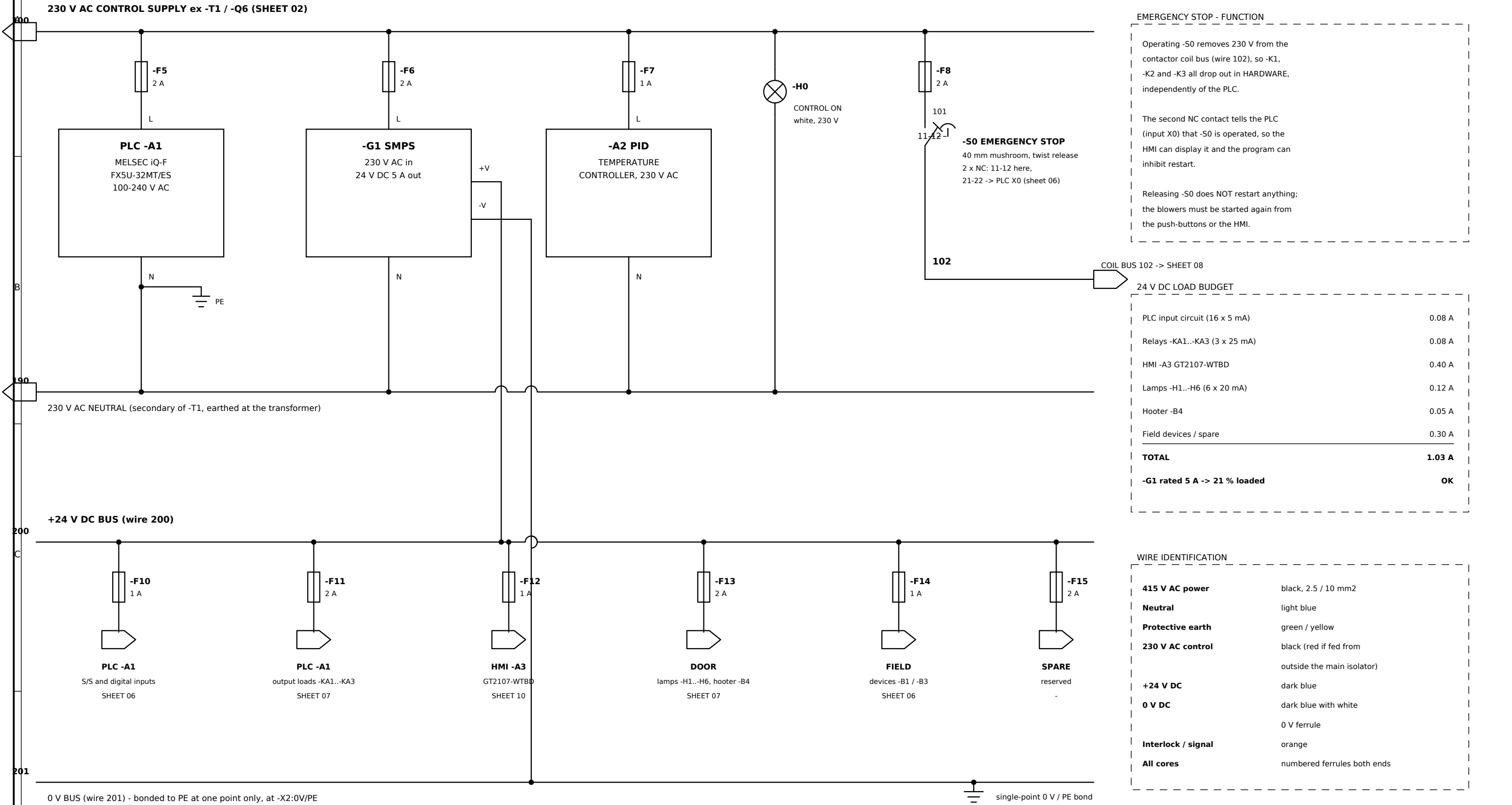
NOTES:

- Elements are connected in STAR with the star point floating. Do not connect the star point to the system neutral - a single element failure would then unbalance the neutral current.
- Q4 is C-curve. The heater is a purely resistive load with a cold-start inrush of about 1.3 x In for a few seconds.
- Element terminal box on the oven is to be wired with high-temperature silicone/glass-fibre insulated cable, rated 180 C minimum, from the gland plate to the element studs.
- Insulation resistance of the element bank to be measured before energising: $> 1 \text{ Mohm}$ at 500 V DC when cold.
- F3 also protects against the loss of one element (current asymmetry) - set it to the measured balanced current.
- Contactor -K3 is an AC-1 rated device switching a resistive load. See the duty block above: the PLC limits the switching frequency so that the mechanical and electrical life of -K3 is not consumed prematurely.

PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE POWER CIRCUIT - HEATER BANK -E1 / -E2 / -E3						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
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CONTROL SUPPLY, EMERGENCY STOP & 24 V DC DISTRIBUTION

230 V AC control circuit ex -T1 / 24 V DC ex -G1



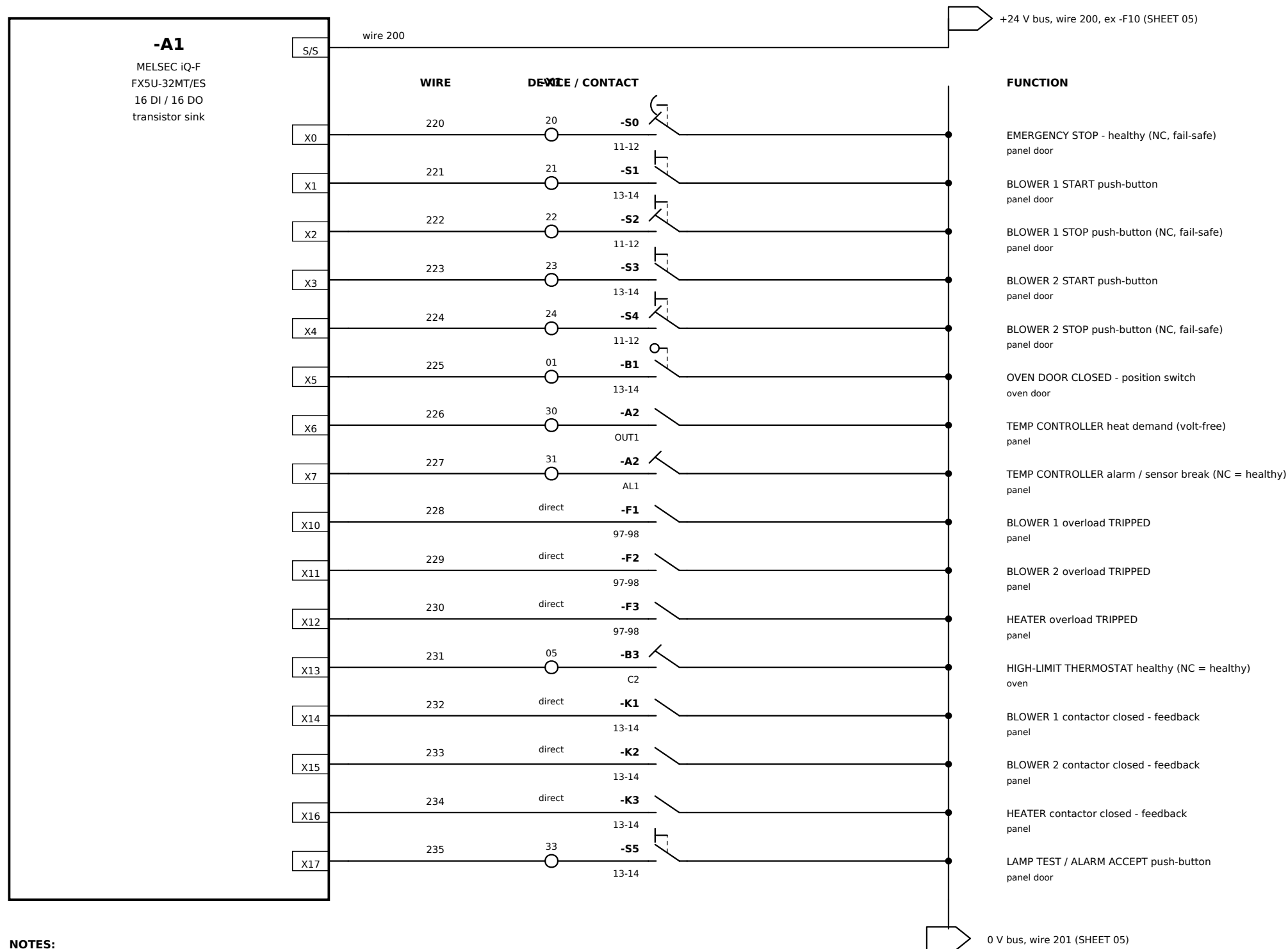
NOTES:

1. -F5 to -F15 are DIN-rail fuse holders with 5 x 20 mm cartridge fuses; the rating is marked beside each holder.
2. The PLC, the SMPS and the PID controller are fed UPSTREAM of the emergency stop so that they stay alive during an E-stop; only the contactor coil bus (wire 102) is interrupted, and that is what removes power from the loads.
3. Wire crossings drawn with a semicircular hop are NOT connected. A junction is shown by a solid dot.

PROJECT				CLIENT / LOCATION		
OVEN CONTROL PANEL - HEATER + 2 BLOWER				-		
SHEET TITLE						
CONTROL SUPPLY, EMERGENCY STOP & 24 V DC DISTRIBUTION						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	05 / 13	1

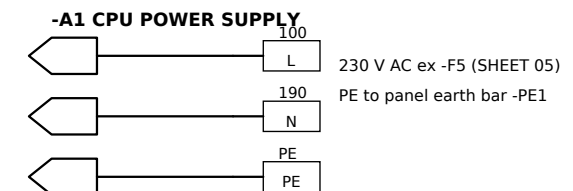
PLC -A1 : CPU SUPPLY & DIGITAL INPUTS X0 - X17

MELSEC iQ-F FX5U-32MT/ES - sink input wiring (S/S = +24 V)



NOTES:

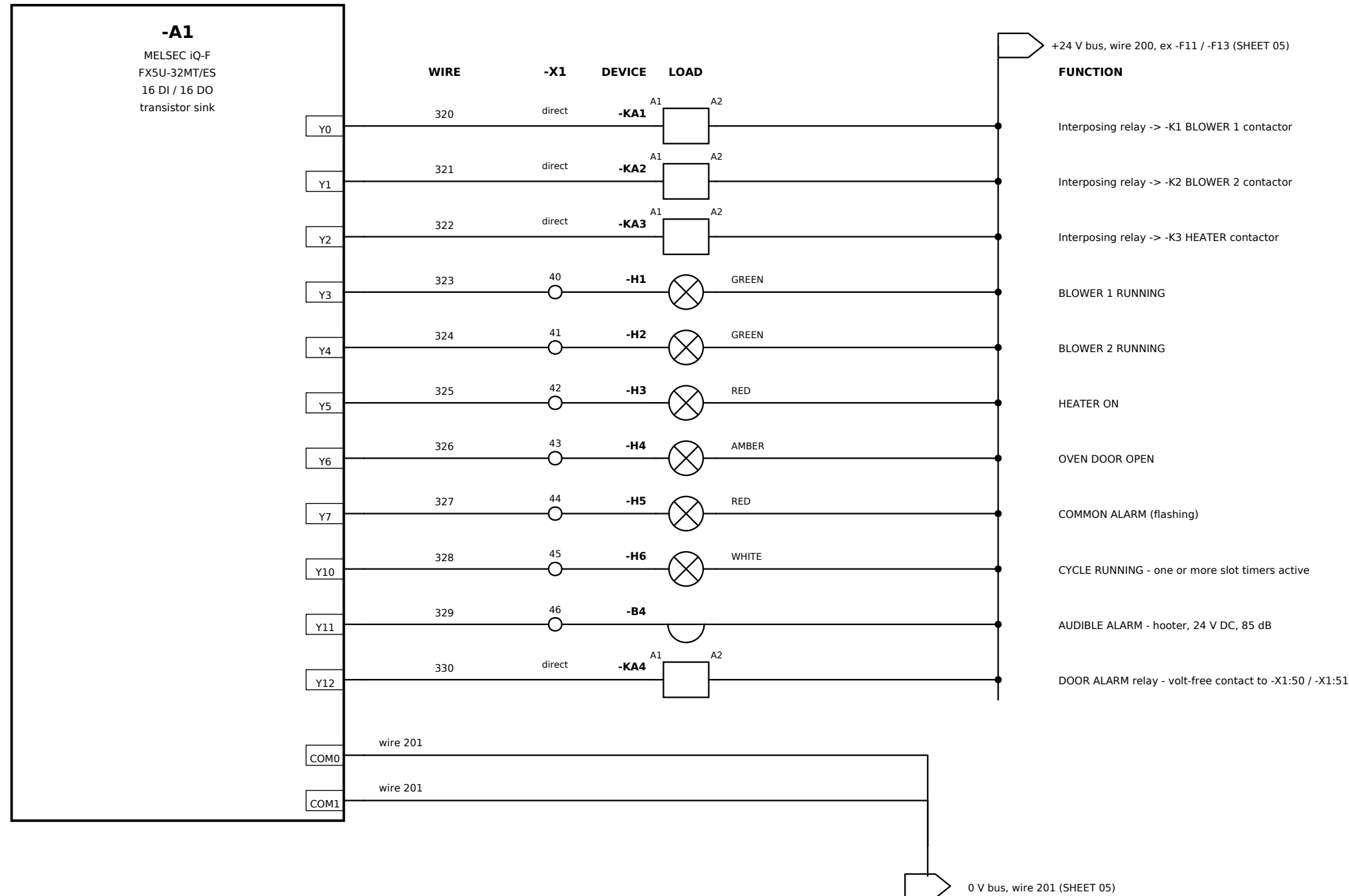
1. SINK input wiring: terminal S/S is tied to +24 V. The internal input circuit sources current out of each X terminal; the field contact completes the circuit to 0 V. A closed field contact = input ON.
2. Fail-safe contacts: -S0 (E-stop), -S2 / -S4 (stop push-buttons) and -B3 (high-limit) are wired normally CLOSED, so the PLC sees the input go OFF on a broken wire, a loose terminal or a device failure, and the program then takes the safe action. The PLC program must therefore treat X0, X2, X4 and X13 as ACTIVE-LOW.
3. Overload trip contacts -F1 / -F2 / -F3 (97-98) are wired normally OPEN so that the input is ON only when that overload has actually tripped - this gives an unambiguous alarm on the HMI.
4. Contactor feedback inputs X14 / X15 / X16 let the program detect a welded or a failed-to-close contactor: if the feedback does not agree with the commanded output within 1 s, a contactor fault alarm is raised.



PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -			
SHEET TITLE PLC -A1 : CPU SUPPLY & DIGITAL INPUTS X0 - X17							
DRAWN CLAUDE		DATE 2026-09-06		DRAWING No.		SHEET REV	
CHECKED -		SCALE NTS		OVN-2026-01		06 / 13 1	

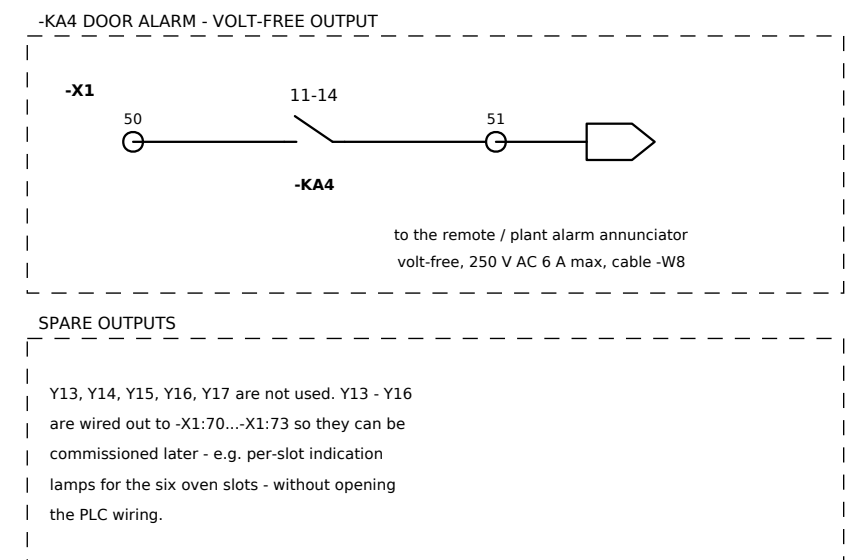
PLC -A1 : DIGITAL OUTPUTS Y0 - Y17

MELSEC iQ-F FX5U-32MT/ES - transistor SINK outputs, 24 V DC, 0.5 A per point



NOTES:

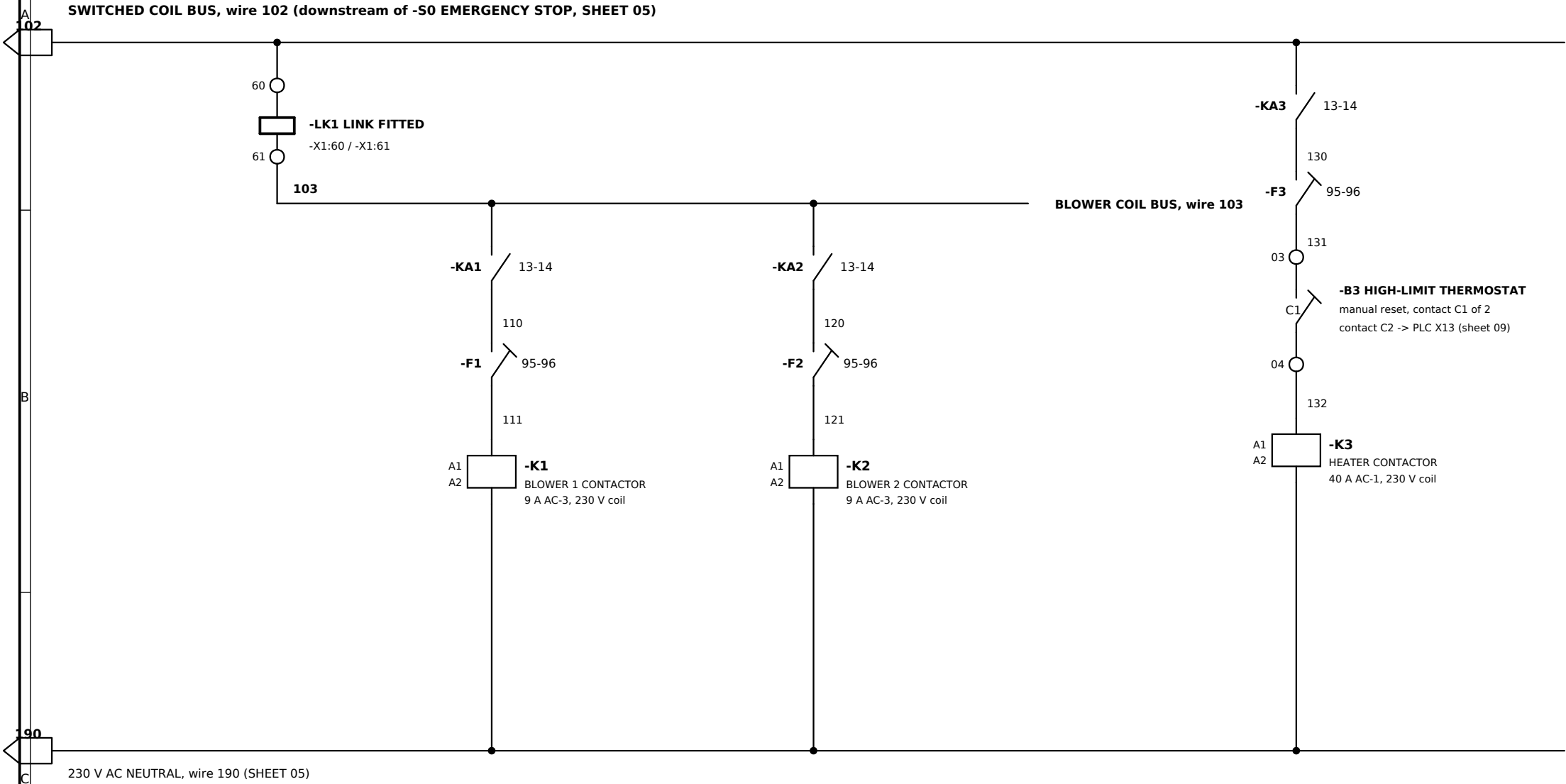
1. Outputs are transistor SINK type. The load is connected between +24 V (wire 200) and the Y terminal; the output transistor completes the circuit to 0 V through COM0 / COM1. Never connect a load between a Y terminal and 0 V - it will be permanently energised.
2. Each output point is rated 0.5 A; each common group is rated 0.8 A total. The loads used here (relay coils 25 mA, LED lamps 20 mA, hooter 50 mA) are well inside both limits.
3. Interposing relays -KA1 / -KA2 / -KA3 have an integral free-wheel diode fitted across the coil. Where a socket-mounted relay without a diode is used, fit a 1N4007 across A1 (+) / A2 (-) to protect the output.
4. Contactor coils are 230 V AC and are NOT driven directly from the PLC - see sheet 08.
5. -H5 COMMON ALARM is flashed by the program (1 s on / 1 s off) while an alarm is unacknowledged and is steady once accepted with -S5. The hooter -B4 silences on -S5 but the lamp stays on until the cause clears.
6. Y12 / -KA4 is the DOOR ALARM output. It is energised by either of the door-signal watchdog conditions - a chattering door switch, or no 'door closed' signal for 10 minutes - and gives a volt-free changeover contact on -X1:50 / -X1:51 for a remote annunciator. The program also flashes -H4 DOOR OPEN for this condition, so a faulty door switch is distinguishable at the panel from a door that is simply open.



PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -			
SHEET TITLE PLC -A1 : DIGITAL OUTPUTS Y0 - Y17							
DRAWN CLAUDE		DATE 2026-09-06		DRAWING No.		SHEET REV	
CHECKED -		SCALE NTS		OVN-2026-01		07 / 13 1	

CONTACTOR COIL CIRCUITS & DOOR-INTERLOCK PROVISION

230 V AC coil circuits fed from the switched coil bus, wire 102



NOTES:

- Interposing relays -KA1 / -KA2 / -KA3 have 24 V DC coils driven from PLC outputs Y0 / Y1 / Y2 (sheet 07). Their volt-free 13-14 contacts switch the 230 V AC contactor coils shown here, so the PLC never carries 230 V and the coil circuit stays independent of the PLC output stage.
- Overload contacts -F1 / -F2 / -F3 (95-96) are in the coil circuits, so an overload trip drops its contactor in HARDWARE. The 97-98 contact of the same relay tells the PLC which overload tripped (sheet 06).
- B3 contact C1 is in the -K3 coil circuit only. An over-temperature therefore removes the heater without any involvement of the PLC or the PID controller. Contact C2 reports the same event to PLC input X13.
- Contactor auxiliary contacts 13-14 of -K1 / -K2 / -K3 report back to PLC inputs X14 / X15 / X16 so the program can detect a welded contactor or one that failed to pull in.
- Fit an RC suppressor across the coil of -K3 to limit the switching transient seen by -KA3.

HOW EACH COIL IS ENERGISED

- K1 / -K2 (blowers)**
- 102 present (E-stop released)
 - AND -LK1 fitted (or the future door switch closed)
 - AND PLC output Y0 / Y1 ON
 - AND that overload not tripped
- K3 (heater)**
- 102 present (E-stop released)
 - AND PLC output Y2 ON
 - AND -F3 not tripped
 - AND -B3 high-limit not tripped
- The door interlock acts on Y0 / Y1 in the PLC program only - it is NOT in the -K3 chain, so the heater stays ON while the door is open.

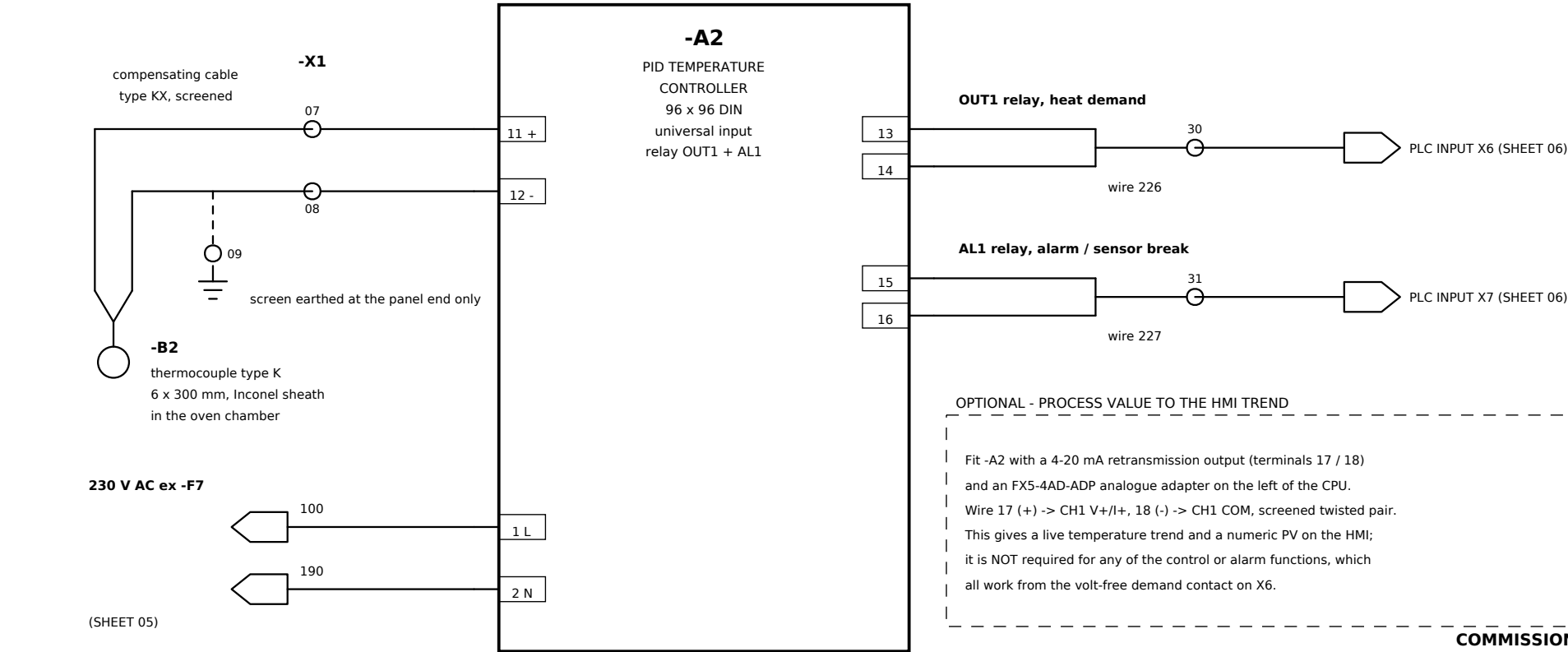
-LK1 UPGRADE TO A HARD-WIRED INTERLOCK

- As instructed, the door switch -B1 goes to PLC input X5 only and the blowers are stopped by the program. The wire link -LK1 between -X1:60 and -X1:61 keeps the blower coil bus permanently made.
- To convert to a hard-wired interlock later:
- Remove the link -LK1.
 - Land the NC contact of a second (safety-rated) door switch -B1.2 on -X1:60 and -X1:61.
 - No other panel change is needed; X5 keeps working for the counters and the HMI status.

PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE CONTACTOR COIL CIRCUITS & DOOR-INTERLOCK PROVISION						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	08 / 13	1

TEMPERATURE CONTROLLER, THERMOCOUPLE & OVER-TEMPERATURE PROTECTION

-A2 PID controller, -B2 type K thermocouple, -B3 independent high-limit thermostat



COMMISSIONING SETTINGS

PARAMETER	SETTING
-A2 input type	K thermocouple, 0 - 400 C
-A2 SV (set point)	as per product recipe (e.g. 180 C)
-A2 control action	reverse (heating), PID
-A2 OUT1 cycle time	20 s minimum (see sheet 04, NOTE 6)
-A2 AL1 type	process high + sensor break, contact NC
-A2 AL1 setting	SV + 15 C
-B3 trip setting	SV + 30 C, or 250 C, whichever is lower
-B3 reset	manual, at the panel, after investigation
PLC heat-up watchdog	60 min, adjustable 10 - 240 min on the HMI
PLC door-signal watchdog	10 min without a 'door closed' signal on X5
PLC door chatter detect	8 or more transitions of X5 within 10 s
PLC history retention	7 days rolling, CSV on the FX5U SD card

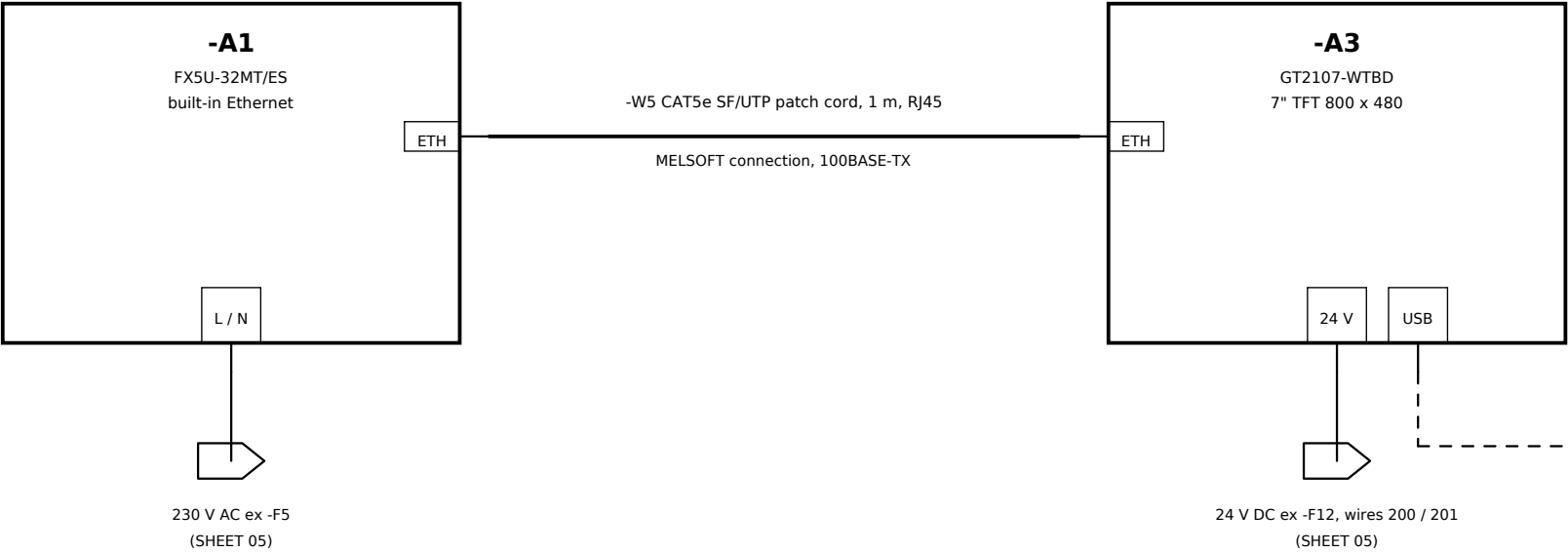
NOTES:

- A2 keeps full local control of the process temperature. The PLC does not compute the temperature loop; it simply repeats the controller's demand contact to the heater contactor and adds the permissives and alarms.
- The demand contact on X6 is volt-free. If the controller supplied is an SSR-drive (12 V DC pulse) type instead of a relay type, fit an interposing relay -KA5 (12 V DC coil) and take its NO contact to X6 - the PLC program is unchanged.
- B2 and the -B3 sensor must be separate sensing elements in separate pockets. A single shared sensor would defeat the whole purpose of the independent over-temperature layer.
- Heat-up watchdog: the PLC starts a timer when the demand contact X6 closes. If X6 stays closed continuously for longer than the watchdog time - i.e. the controller never gets to set point and drops out - the alarm 'FAILURE TO REACH SETPOINT' is raised on the HMI and the hooter sounds. The heater is not tripped by this alarm; it is an operator warning, since the real protection against over-temperature is -B3.

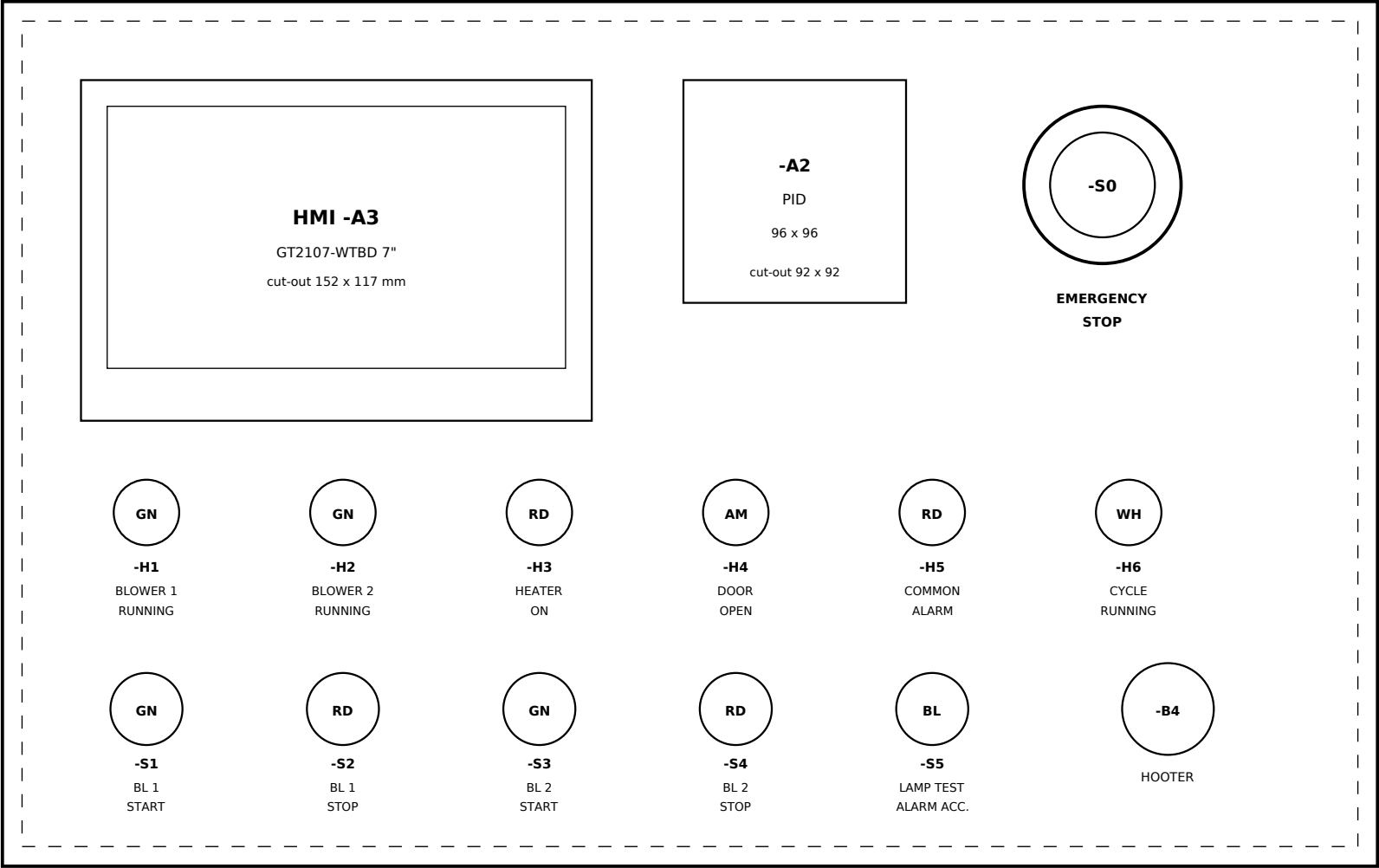
PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE TEMPERATURE CONTROLLER, THERMOCOUPLE & OVER-TEMPERATURE PROTECTION						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	09 / 13	1

HMI -A3, ETHERNET NETWORK & PANEL DOOR ARRANGEMENT

GOT2000 GT2107-WTBD <-> MELSEC iQ-F FX5U-32MT/ES, 100BASE-TX



PANEL DOOR - EXTERNAL ARRANGEMENT (view on front, 800 x 600 mm door)



NETWORK ADDRESSING

DEVICE	IP ADDRESS	NOTES
-A1 FX5U	192.168.3.250	built-in port, default
-A3 GT2107	192.168.3.18	GOT default
Engineering PC	192.168.3.10	GX Works3 / GT Designer3
Subnet mask	255.255.255.0	-
Gateway	not used	isolated machine network

HMI SCREEN STRUCTURE (see the HMI project for detail)

SCREEN	TITLE	SEC. LEVEL
B-1000	HOME / OVERVIEW	0
B-1100	SLOT TIMERS 1-6	0
B-1200	COUNTERS - door, lots	0
B-1300	ALARMS - live & history	0
B-1400	MANUAL TEST - blowers, heater	2
B-1500	SETTINGS - watchdog, presets	2
B-1600	PASSWORD / LOGIN	0
B-1700	TIMER EARLY RESET	2
B-1800	COUNTER RESET	2
B-1900	HISTORY - log view & USB export	0

Security level 0 = operator, no login required.
Level 2 = maintenance technician, password protected.

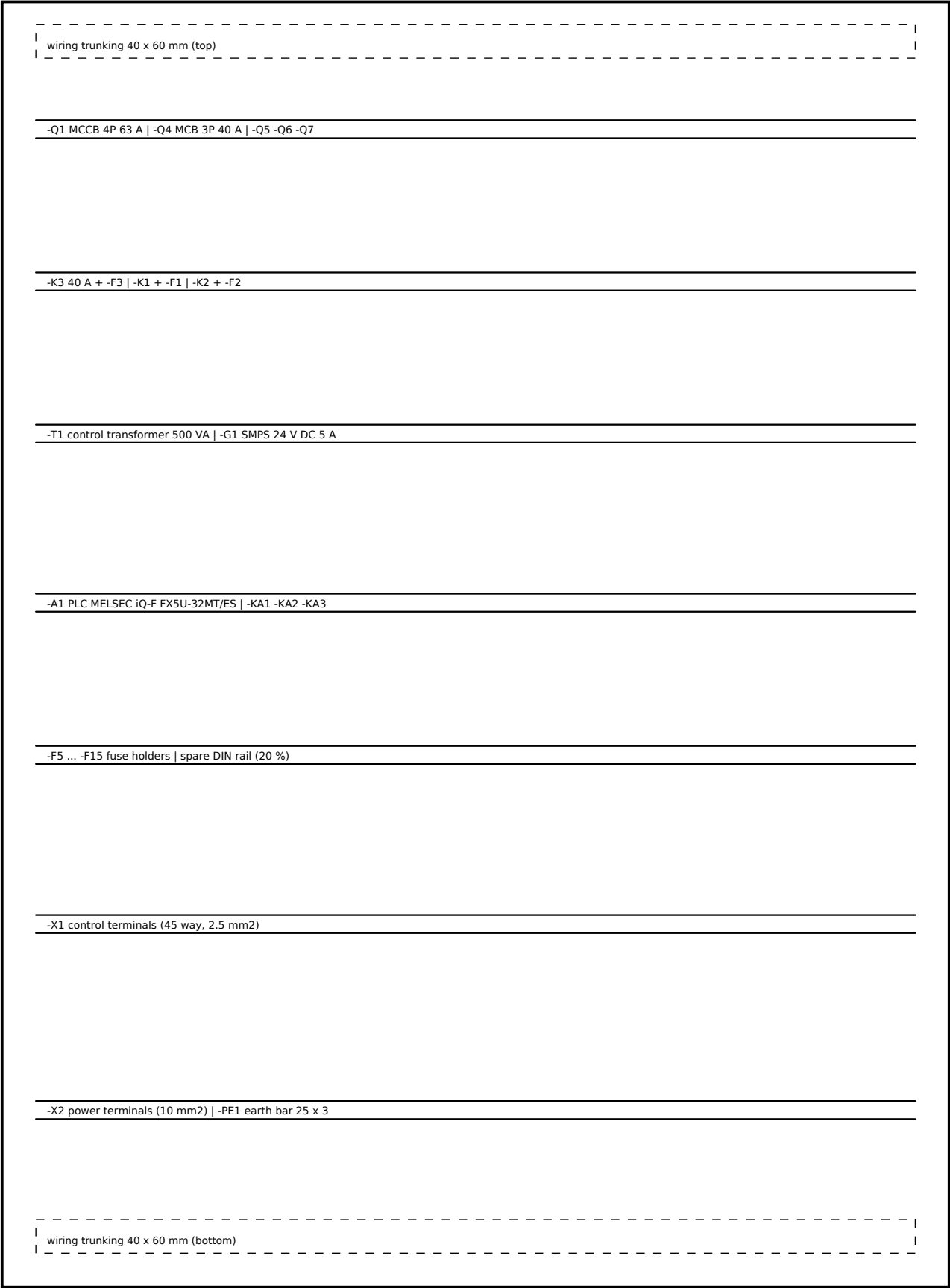
- NOTES:**
- The HMI is powered from -F12 and is NOT interrupted by the emergency stop, so alarms and the timer values stay visible.
 - Slot timer values, counters and the alarm history are held in PLC latched memory (see the PLC program device list), not in the HMI, so they survive a power failure and an HMI replacement.
 - The GOT battery (GT11-50BAT) is fitted for the real-time clock used by the alarm and the operation-log time stamps.
 - The 7-day alarm and event history is written as CSV to the SD card in the FX5U, using the CPU data-logging function, and is exported to a USB stick from screen B-1900.
 - Door cut-outs to be made before painting; use the gaskets supplied to keep IP54.

PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE HMI -A3, ETHERNET NETWORK & PANEL DOOR ARRANGEMENT						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	10 / 13	1

PANEL GENERAL ARRANGEMENT & GLAND PLATE

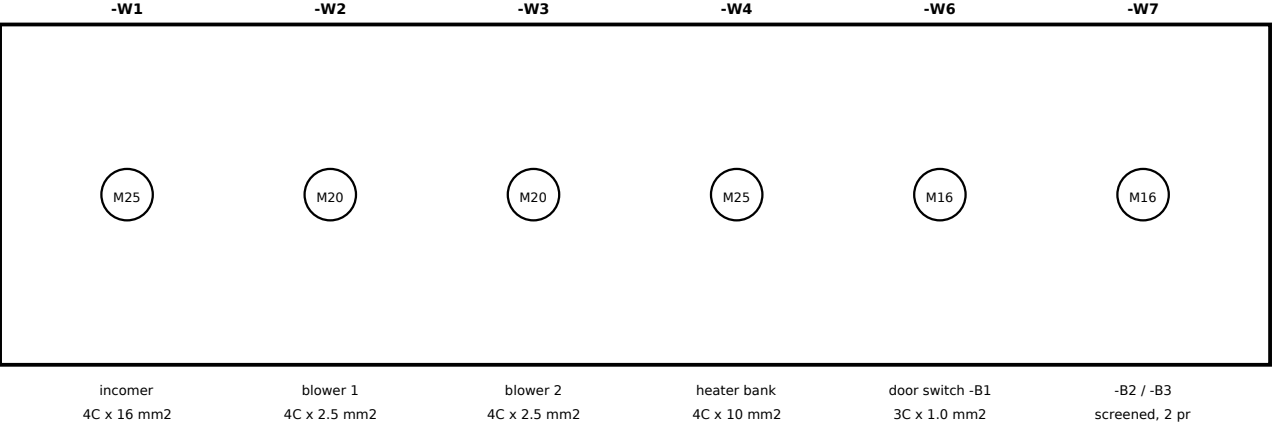
Wall-mounted enclosure 800 H x 600 W x 250 D mm, IP54, RAL 7035

MOUNTING PLATE - INTERNAL LAYOUT (scale approx. 1:3.3)



mounting plate 560 x 760 mm

GLAND PLATE - BOTTOM OF ENCLOSURE



ENCLOSURE & CONSTRUCTION

ITEM	SPECIFICATION
Enclosure	800 H x 600 W x 250 D mm, mild steel 1.5 mm, IP54
Finish	powder coated RAL 7035, plate RAL 9003
Mounting	wall mounted, 4 x M8 bolts, bottom of panel 1000 mm AFFL
Door	single door, 3-point lock, earthed with 4 mm2 braid
Ventilation	filter fan 100 m3/h + exhaust filter, thermostat set 35 C
Cable entry	bottom gland plate, removable, glands per the detail above
Internal wiring	in slotted trunking, minimum 60 % spare capacity
Segregation	415 V power and 24 V DC control in separate trunking runs
Earthing	copper earth bar -PE1 25 x 3 mm, all doors and plates bonded
Labelling	engraved traffolyte for devices, printed ferrules for cores
Spare capacity	20 % free DIN rail and 4 spare terminals per strip
Tests	IR 500 V DC, flash 1.5 kV 1 min, functional test to sheet 13

INSTALLATION REQUIREMENTS

- Keep 800 mm clear working space in front of the panel (IEC 60204-1).
- The panel must not be mounted on the oven skin or within 1 m of it - radiated heat will drive the internal temperature above the PLC rating (55 C). Mount on the adjacent wall or on a free-standing frame.
- Thermocouple and 4-20 mA cables to be run at least 300 mm away from the motor and heater power cables, crossing at right angles only.
- The oven body, the element housing and the blower frames must all be bonded to -PE1. Measure the bond resistance: < 0.1 ohm.

NOTES:

- Layout shown is indicative; the panel builder may rearrange within the same enclosure provided the segregation and clearance requirements are kept.
- Fit an internal panel light and a 6 A socket outlet fed from -Q7 (RCBO).
- Fit a door-mounted pocket for this drawing set and the O&M manual.

PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE PANEL GENERAL ARRANGEMENT & GLAND PLATE						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	11 / 13	1

TERMINAL STRIP SCHEDULE -X1 / -X2

All field and door-loom wiring terminates here; panel-plate devices are wired direct

-X1 CONTROL TERMINALS (2.5 mm2)

TERM	WIRE	DEVICE	DESCRIPTION	SH.
01	225	-B1:13	Oven door position switch - contact 13	06
02	201	-B1:14	Oven door position switch - contact 14 (0 V)	06
03	102	-B3:C1	High-limit thermostat contact 1 - from -KA3	08
04	132	-B3:C1	High-limit thermostat contact 1 - to -K3 A1	08
05	231	-B3:C2	High-limit thermostat contact 2 - to PLC X13	06
06	201	-B3:C2	High-limit thermostat contact 2 - 0 V return	06
07	TC+	-B2:+	Thermocouple type K, positive (green)	09
08	TC-	-B2:-	Thermocouple type K, negative (white)	09
09	SCR	-B2 scr	Thermocouple screen - earthed at panel only	09
10	PE	oven	Oven body bonding conductor	04
20	220	-S0:11	Emergency stop, contact 11-12 -> PLC X0	06
21	221	-S1:13	Blower 1 START push-button -> PLC X1	06
22	222	-S2:11	Blower 1 STOP push-button -> PLC X2	06
23	223	-S3:13	Blower 2 START push-button -> PLC X3	06
24	224	-S4:11	Blower 2 STOP push-button -> PLC X4	06
30	226	-A2:OUT1	PID controller heat demand -> PLC X6	09
31	227	-A2:AL1	PID controller alarm -> PLC X7	09
33	235	-S5:13	Lamp test / alarm accept -> PLC X17	06
34	201	0 V	0 V common for door-mounted push-buttons	06

-X2 POWER TERMINALS (10 mm2 + PE)

TERM	WIRE	DEVICE	DESCRIPTION
01	11	-M1 U1	Blower 1 motor, phase U
02	12	-M1 V1	Blower 1 motor, phase V
03	13	-M1 W1	Blower 1 motor, phase W
04	21	-M2 U1	Blower 2 motor, phase U
05	22	-M2 V1	Blower 2 motor, phase V
06	23	-M2 W1	Blower 2 motor, phase W
10	31	-E1	Heater element 1, line
11	32	-E2	Heater element 2, line
12	33	-E3	Heater element 3, line
PE	PE	PE bar	Protective earth - motors, oven body, gland plate

-X1 CONTROL TERMINALS (continued)

TERM	WIRE	DEVICE	DESCRIPTION	SH.
35	201	0 V	0 V common for door-mounted push-buttons	06
40	323	-H1	Lamp BLOWER 1 RUNNING (green) <- Y3	07
41	324	-H2	Lamp BLOWER 2 RUNNING (green) <- Y4	07
42	325	-H3	Lamp HEATER ON (red) <- Y5	07
43	326	-H4	Lamp DOOR OPEN (amber) <- Y6	07
44	327	-H5	Lamp COMMON ALARM (red) <- Y7	07
45	328	-H6	Lamp CYCLE RUNNING (white) <- Y10	07
46	329	-B4	Hooter 24 V DC <- Y11	07
47	200	+24 V	+24 V common for door-mounted lamps / hooter	07
48	200	+24 V	+24 V common for door-mounted lamps / hooter	07
50	VF1	-KA4:11	DOOR ALARM volt-free contact - remote / plant alarm	07
51	VF2	-KA4:14	DOOR ALARM volt-free contact - remote / plant alarm	07
60	102	LINK	PROVISION - hard-wired door interlock, link -LK1 fitted	08
61	103	LINK	PROVISION - hard-wired door interlock, link -LK1 fitted	08
70	-	-	SPARE	-
71	-	-	SPARE	-
72	-	-	SPARE	-
73	-	-	SPARE	-

TERMINAL STRIP CONVENTIONS

1. -X1 is a single strip in terminal-number order with a marked end plate at each end.
2. Terminals 01-10 are the OVEN field devices, 20-35 the panel DOOR loom, 40-48 the door indication loom, 60-61 the door-interlock provision and 70-73 spare.
3. Every core carries a printed ferrule with the wire number shown, at BOTH ends.
4. A grey terminal is used for all signals, light blue for 0 V and green/yellow for PE.
5. No more than TWO conductors in any one terminal; use a jumper comb for commons.
6. -X1:60 / -X1:61 carry the link -LK1. Do not remove the link unless a hard-wired door interlock is being fitted - the blowers will not run with the link removed.
7. Spare terminals 70-73 are pre-wired to PLC outputs Y12-Y15 for future use.

NOTES:

1. Wire numbers follow the scheme: 1xx = 230 V AC control, 2xx = 24 V DC inputs, 3xx = 24 V DC outputs, and power cores are numbered by feeder (1L1, 2L1, 31 ...).
2. Check every ferrule against this schedule at pre-delivery inspection and again after installation; a mis-numbered core is the single most common cause of a failed FAT.

PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE TERMINAL STRIP SCHEDULE -X1 / -X2						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	12 / 13	1

I/O SCHEDULE & BILL OF MATERIALS

MELSEC iQ-F FX5U-32MT/ES - 16 digital inputs, 16 digital outputs

DIGITAL INPUTS

ADDR	WIRE	-X1	DEVICE	CONTACT	DESCRIPTION
X0	220	20	-S0	11-12	EMERGENCY STOP - healthy (NC, fail-safe)
X1	221	21	-S1	13-14	BLOWER 1 START push-button
X2	222	22	-S2	11-12	BLOWER 1 STOP push-button (NC, fail-safe)
X3	223	23	-S3	13-14	BLOWER 2 START push-button
X4	224	24	-S4	11-12	BLOWER 2 STOP push-button (NC, fail-safe)
X5	225	01	-B1	13-14	OVEN DOOR CLOSED - position switch
X6	226	30	-A2	OUT1	TEMP CONTROLLER heat demand (volt-free)
X7	227	31	-A2	AL1	TEMP CONTROLLER alarm / sensor break (NC = healthy)
X10	228	-	-F1	97-98	BLOWER 1 overload TRIPPED
X11	229	-	-F2	97-98	BLOWER 2 overload TRIPPED
X12	230	-	-F3	97-98	HEATER overload TRIPPED
X13	231	05	-B3	C2	HIGH-LIMIT THERMOSTAT healthy (NC = healthy)
X14	232	-	-K1	13-14	BLOWER 1 contactor closed - feedback
X15	233	-	-K2	13-14	BLOWER 2 contactor closed - feedback
X16	234	-	-K3	13-14	HEATER contactor closed - feedback
X17	235	33	-S5	13-14	LAMP TEST / ALARM ACCEPT push-button

DIGITAL OUTPUTS

ADDR	WIRE	-X1	DEVICE	COLOUR	DESCRIPTION
Y0	320	-	-KA1	-	Interposing relay -> -K1 BLOWER 1 contactor
Y1	321	-	-KA2	-	Interposing relay -> -K2 BLOWER 2 contactor
Y2	322	-	-KA3	-	Interposing relay -> -K3 HEATER contactor
Y3	323	40	-H1	GREEN	BLOWER 1 RUNNING
Y4	324	41	-H2	GREEN	BLOWER 2 RUNNING
Y5	325	42	-H3	RED	HEATER ON
Y6	326	43	-H4	AMBER	OVEN DOOR OPEN
Y7	327	44	-H5	RED	COMMON ALARM (flashing)
Y10	328	45	-H6	WHITE	CYCLE RUNNING - one or more slot timers active
Y11	329	46	-B4	-	AUDIBLE ALARM - hooter, 24 V DC, 85 dB
Y12	330	-	-KA4	-	DOOR ALARM relay - volt-free contact to -X1:50 / -X1:51
Y13	-	-	-	-	SPARE
Y14	-	-	-	-	SPARE
Y15	-	-	-	-	SPARE
Y16	-	-	-	-	SPARE
Y17	-	-	-	-	SPARE

NOTES:

1. I/O count used: 16 of 16 digital inputs (0 spare), 11 of 16 digital outputs (5 spare).
- The input card is full. If any further input is needed - for example a second door switch, a per-slot load sensor or an airflow proving switch - add an FX5-16EX/ES input extension block on the right of the CPU; no rewiring of the existing inputs is needed.
2. Quantities in the bill of materials exclude consumables (ferrules, cable ties, glands, labels, wire).
3. Equivalent devices from another manufacturer may be substituted provided the ratings, the utilisation category and the terminal arrangement are the same, and the substitution is recorded on the as-built.
4. All settings marked 'set to ...' on sheets 03, 04 and 09 must be recorded on the commissioning sheet.

BILL OF MATERIALS

#	TAG	DESCRIPTION	TYPE / MAKE	QTY
1	-Q1	MCCB 4P 63 A, Icu 25 kA, adj. 40-63 A	Mitsubishi NF63-CV 4P 63A	1
2	-Q2/-Q3	Motor protection circuit breaker 3P 2.5-4 A	Schneider GV2ME08	2
3	-Q4	MCB 3P 40 A C-curve 10 kA	Mitsubishi BH-DN 3P C40	1
4	-Q5	MCB 2P 6 A C-curve	Mitsubishi BH-DN 2P C6	1
5	-Q6	MCB 1P 4 A C-curve	Mitsubishi BH-DN 1P C4	1
6	-Q7	RCBO 2P 6 A, 30 mA	Mitsubishi BV-DN 2P C6	1
7	-K1/-K2	Contactor 9 A AC-3, 230 V AC coil, 1NO+1NC aux	Mitsubishi S-T10 + UT-AX	2
8	-K3	Contactor 40 A AC-1, 230 V AC coil, 1NO+1NC aux	Mitsubishi S-T21 + UT-AX	1
9	-F1/-F2	Thermal overload relay 2.5-4 A, class 10A	Mitsubishi TH-T18 3.6A	2
10	-F3	Thermal overload relay 23-32 A, class 10A	Mitsubishi TH-T25 29A	1
11	-T1	Control transformer 415/230 V, 500 VA, screened	IEC 61558-2-6	1
12	-G1	SMPS 230 V AC / 24 V DC, 5 A, 120 W, DIN rail	Mean Well NDR-120-24	1
13	-A1	PLC CPU MELSEC iQ-F FX5U-32MT/ES	Mitsubishi FX5U-32MT/ES	1
14	-A3	HMI GOT2000 GT2107-WTBD 7" TFT 800x480	Mitsubishi GT2107-WTBD	1
15	-A2	PID temperature controller 96x96, universal input,	Yokogawa UT35A / equiv.	1
16	-B2	Thermocouple type K, 6 mm x 300 mm, Inconel sheath	with compensating cable	1
17	-B3	Safety high-limit thermostat, 2 x NC, manual reset	e.g. JUMO STB, 0-400 C	1
18	-B1	Position switch, roller lever, 1NO+1NC, IP67	e.g. Schneider XCKN	1
19	-KA1...KA4	Interposing relay 24 V DC coil, 1 c/o 6 A, socket	e.g. Finder 40.52 + socket	4
20	-S0	Emergency stop, 40 mm mushroom, twist release, 2NC	22 mm, IP66	1
21	-S1...S4	Push-button 22 mm, flush, 1NO or 1NC	green x2 / red x2	4
22	-S5	Push-button 22 mm, blue, 1NO	lamp test / alarm accept	1
23	-H1...H6	Pilot lamp 22 mm LED 24 V DC	2 green, 2 red, 1 amber, 1 white	6
24	-B4	Alarm hooter 24 V DC, 85 dB	22 mm or surface	1
25	-X1	Terminal block 2.5 mm2 screw / spring, grey	with end plates and markers	45
26	-X2	Terminal block 10 mm2, grey + green/yellow PE	with end plates and markers	12
27	-W5	Ethernet patch cord CAT5e SF/UTP, 1 m, RJ45	industrial rated	1
28	-	Enclosure 800 x 600 x 250 mm, IP54, RAL 7035	wall mounting, with gland plate	1
29	-	Panel light, 6 A socket, thermostat + filter fan 100 m3/h	-	1
30	-A1	SD memory card 2 GB for the FX5U data logging function	Mitsubishi NZ1MEM-2GBSD	1
31	-A3	GOT battery for the real-time clock and alarm history	Mitsubishi GT11-50BAT	1
32	-	DIN rail, trunking 40x60, earthing bar, labels, ferrules	consumables	1

PROJECT OVEN CONTROL PANEL - HEATER + 2 BLOWER				CLIENT / LOCATION -		
SHEET TITLE I/O SCHEDULE & BILL OF MATERIALS						
DRAWN	CLAUDE	DATE	2026-09-06	DRAWING No.	SHEET	REV
CHECKED	-	SCALE	NTS	OVN-2026-01	13 / 13	1